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AI-enabled innovation and firm growth in the Asia–Pacific region: Multilevel and experimental evidence on regulatory climate

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ABSTRACT

The present research investigates how AI-enabled innovation translates into firm productivity and economic growth under varying regulatory conditions. The primary goal of this study is to explain when and how firms are able to convert AI-enabled innovation into measurable productivity gains and sustained growth, and under which regulatory environments these effects are strengthened or constrained. Drawing on institutional and contingency perspectives, two complementary studies are conducted to test a moderated mediation model. Study 1 employs multilevel survey data from 420 firms across multiple regions, analyzed using hierarchical linear modeling. Results suggest that AI-enabled innovation is associated with firm growth both directly and indirectly through productivity gains, with the strength of the innovation–productivity link contingent on the regional regulatory climate. Study 2 adopts an experimental vignette design with 360 managerial participants to provide causal evidence. Regulatory clarity and supportive enforcement are manipulated, and results indicate that clear and supportive environments are associated with higher levels of intended AI investment, productivity expectations, deployment speed, and growth projections. Mediation analyses confirm that productivity expectations transmit the effects of regulatory conditions to growth, and moderated mediation findings reveal that the productivity pathway is strongest when both clarity and supportiveness are present. Together, these studies provide convergent evidence that institutional environments may amplify the benefits of AI adoption, offering actionable implications for innovation and regulatory policy by showing that environmental policies emphasizing regulatory clarity, consistent enforcement, and experimentation-friendly frameworks are likely to play an important role in enabling firms to translate AI investments into productivity improvements and economic growth.

1. Introduction

Artificial intelligence (AI) is increasingly recognized as a transformative force reshaping industries, altering competitive dynamics, and influencing broader patterns of economic growth (Trabelsi, 2024). According to recent estimates, AI-driven technologies are expected to contribute trillions of dollars to global GDP within the next decade, highlighting their potential to fundamentally reconfigure how firms create value and sustain growth (Gondauri, 2025). Yet despite this promise, a persistent puzzle remains: while some firms successfully leverage AI to enhance productivity and stimulate growth, others struggle to realize measurable returns from similar investments. Recent empirical evidence suggests that AI adoption alone does not guarantee uniform economic benefits, highlighting substantial heterogeneity in firm-level outcomes (Babina et al., 2025; Frimpong, 2025). Understanding why the outcomes of AI-enabled innovation vary so widely is critical for both scholars and practitioners concerned with the intersection of technology adoption, firm performance, and institutional environments (see Figs. 1 and 2).

Prior research has established that innovation, broadly conceived, is a key driver of productivity improvements and economic growth (Albis Salas et al., 2023; Shen et al., 2025). Within this stream, AI-enabled innovation represents a particularly powerful mechanism because of its capacity to automate complex processes, enable predictive analytics, and generate new business models (Yi & Ayangbah, 2024). Empirical studies have begun to demonstrate positive associations between AI adoption and firm-level outcomes such as efficiency, decision quality, and competitive advantage (Giachino et al., 2024; Rammer et al., 2021;

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